27. Weather and calendar — forecasting, telling time, planning the year

Oracle OS guide. Manuals in the library: "Weather Proverbs and Paradoxes", "The Weather and Practical Methods of Forecasting", "Weather Lore", "Clouds and Weather" (Gutenberg), Boy Scouts Handbook, navigation (stars).

Reading clouds

- **Cirrus** (high wisps) thickening into **cirrostratus** (veil, halo round sun/moon) → rain in 12–24 h (warm front).
- Grey **altostratus** lowering → steady rain on the way.
- Fair-weather **cumulus** (cotton, flat base) in the morning = fine; growing into towers in the afternoon (**cumulonimbus**, anvil top) = thunder, hail, gusts.
- Low **stratus**/fog in the morning that lifts = good day.
- "Mackerel sky" (**altocumulus** scales) on a summer morning = afternoon thunderstorms.

Other signs

- Wind veering clockwise (northern hemisphere) and strengthening → front passing; moist south/south-westerly wind → rain in western Europe.
- "Red sky at night, shepherd's delight; red sky in the morning, shepherd's warning" (works with weather arriving from the west).
- Morning dew/frost = clear night, no rain imminent. No dew on a clear night = wind or cloud coming.
- Smoke sinking and spreading, sounds carrying, strong smells, birds flying low, insects biting → falling pressure, rain.
- **Thunderstorms**: seconds between flash and thunder ÷ 3 = km. Under 30 s, take shelter (not under lone trees; crouch with feet together if exposed).
- **Improvised barometer**: jar of water with an inverted straw/tube, or a balloon stretched over a jar with a pointer — the trend matters more than the value: falling fast = bad weather.
- **Frost**: clear, still, dry nights — protect plants (cover, water in the late afternoon).

Telling the time

- **Sundial**: vertical stake, mark the shadow hourly for a day; shortest shadow = solar noon. For accuracy: gnomon tilted at an angle = latitude, pointing at the pole.
- **Hand on the horizon**: each finger at arm's length ≈ 15 min of sun before sunset.
- **Night**: stars turn 15°/hour round the Pole Star; the Plough works like a clock hand. Moon phases (below).
- Marked candles, sand-glasses, water clocks for intervals.

Calendar

- **Solstices** (sun rises at its most northern/southern point on the horizon — mark it with aligned stones/stakes): ~21 Jun and ~21 Dec. **Equinoxes** (sun rises due east, day = night): ~20 Mar and ~22 Sep.
- Year = 365¼ days → leap year every 4 years (except century years not divisible by 400). Keep a **tally stick** or written calendar, with a keeper and a copy.
- **Moon**: 29.5-day cycle. Full moon rises at sunset; a waxing moon is seen in the evening in the west (lit on the right in the northern hemisphere). Useful for counting months and tides (spring tides at new and full moon).
- **Day of the week** for any date: keep a printed perpetual calendar (Print pack).

Farming calendar (temperate northern hemisphere — adjust locally)

- **Jan–Feb**: plan, repair tools, prune fruit trees, sow under cover.
- **Mar–Apr**: sow peas, broad beans, onions, potatoes; prepare beds; lambing/kidding.
- **May–Jun**: after the last frosts, sow beans, squash, maize; shearing; haymaking.
- **Jul–Aug**: grain harvest; dry and store seed; preserving.
- **Sep–Oct**: lift potatoes, apples, nuts; sow winter wheat/rye; slaughter and salting.
- **Nov–Dec**: manure, store, crafts, teaching.
- Keep a **field diary**: dates of frosts, sowings, harvests and rain every year — in 5–10 years it gives a reliable local calendar.